Protogold and the 0-255 scale

Panama Blot Analyzer 0.2.3 | Quick walkthrough

1. View the matching simulated scans

Open **experiments/Protogold-walkthrough/02-aligned-and-measured.panama**. In **01 Images & lanes**, click **Western** or **Protogold**. Both scans are embedded in this experiment. The original 18-sample warped-groups test also contains both channels.

Western shows antibody-binding signal. Protogold shows protein landmarks that locate the same migration positions across lanes. Here, 17 interlane strips serve 18 sample lanes; the final strip is shared. This pair is entirely synthetic and has known migration shifts.

2. See what alignment changes

Open **01-before-alignment.panama** to practice. In **02 Alignment**, inspect the eight matched protein landmarks, then click **Apply alignment**. Panama uses the resulting coordinate mapping on both protein and Western profiles. It does not subtract protein intensity from Western intensity. See **Protogold-explained.png** beside the experiment for the before/after illustration.

3. Subtract background separately

In **03 Baseline**, choose Constant, Level **0.06**, check Apply to all lanes and Apply baseline. This is the known background in these synthetic images; it is not a general setting for real membranes. It corresponds to **15.3** in 0-255 signal units. Load the supplied **sections.json** through Sections > Template > Load template.

4. Select the new measurement

In **05 Measurements**, choose **Mean signal (0-255)**. The same option is available under Measure in Analysis and is saved in Excel exports and experiments. A corrected normalized mean of **0.50** becomes **127.5**; the retained Nirvana peak-value formula gives **50** on that normalized input.

This is a change of signal units, not conversion of the image to 8-bit pixels. Individual 8-bit pixels are integers, but means can be decimals: 100 and 101 average to 100.5. Background correction and interpolation also produce fractions. Dark-band mode uses 0 for white and 255 for black before correction; Bright-band mode uses the opposite signal direction. Corrected values are not capped at 255. Optical density is a different, logarithmic quantity and is not converted by this option.

For imported numeric profiles, declare Linear signal 0-1 or Linear signal 0-255 during import, or under Sample > Blot units. Unspecified units remain unavailable rather than being guessed. Existing Area, Mean and Nirvana peak value are unchanged.

5. Interpret the analysis plots

K-means: clusters are fitted using all retained sections; colors show membership on two PCA axes. **LDA:** LD axes use known group labels to separate classes; two groups provide one LD axis. Validation seed changes held-out splits and may change accuracy, but leaves plotted LD coordinates unchanged. Separation in a plot alone does not establish predictive validity.

The published Vale figure has no matching protein scan available. The simulated pair teaches the procedure; it is not a reconstruction of that missing experimental scan.